



< Previous

 ✓

 ✓



Next >

10.1.3 What you will learn

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This week you explore practical methods for finding all eigenvalues and eigenvectors of a Hermitian matrix, building on the insights regarding the Power Method that you discovered last week.

Upon completion of this week, you should be able to

- Formulate and analyze subspace iteration methods.
- Expose the relationship between subspace iteration and simple QR algorithms.
- Accelerate the convergence of QR algorithms by shifting the spectrum of the matrix.
- Lower the cost of QR algorithms by first reducing a Hermitian matrix to tridiagonal form.
- Cast all computation for computing the eigenvalues and eigenvectors of a Hermitian matrix in terms of unitary similarity transformations, yielding the Francis Implicit QR Step.
- Exploit a block diagonal structure of a matrix to deflate the Hermitian eigenvalue problem into smaller subproblems.
- Combine all these insights into a practical algorithm.

< Previous

Next >

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